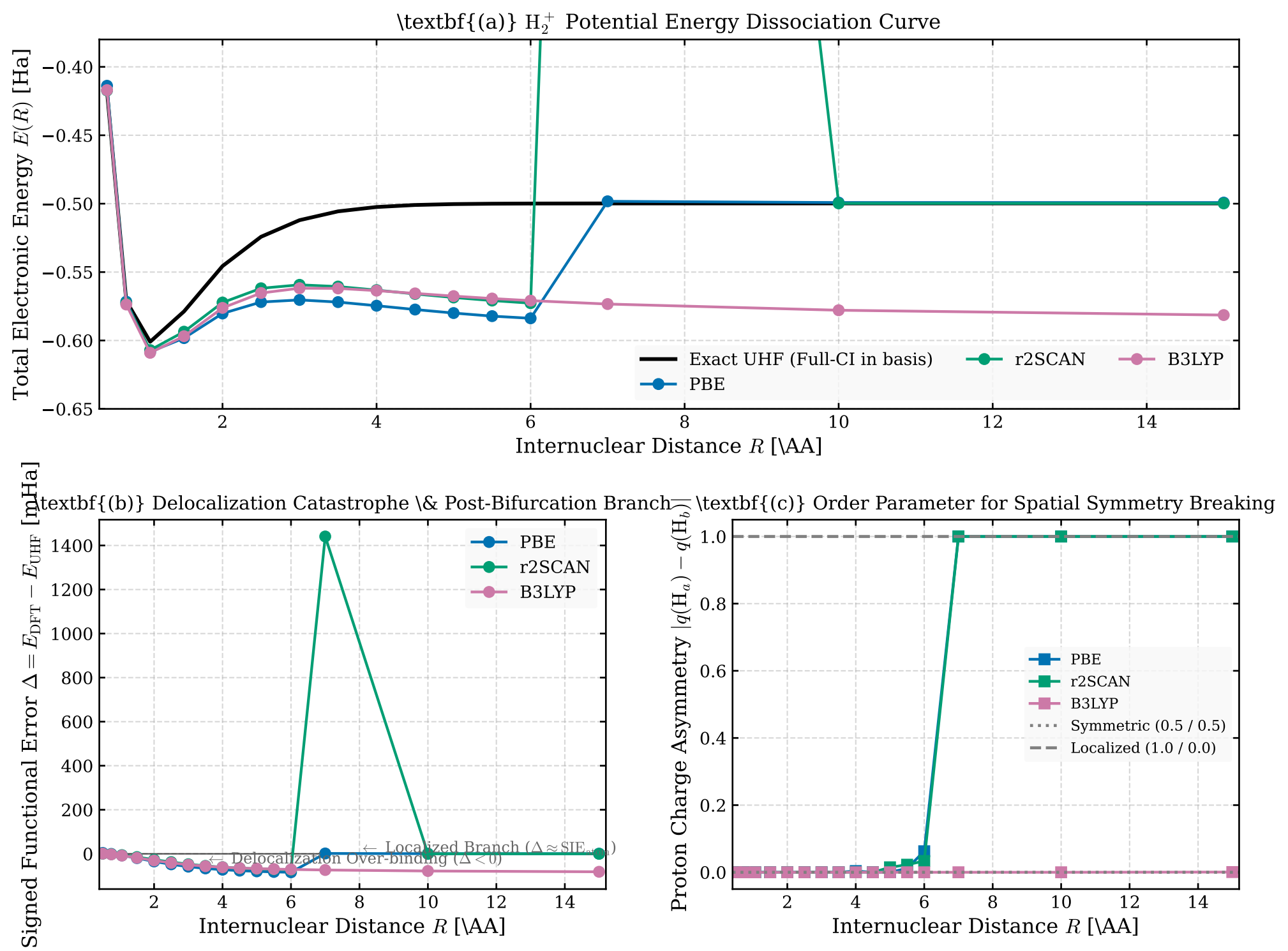
\textbf{(a)} H_2^+ Potential Energy Dissociation Curve
Total Electronic Energy $E(R)$ [Ha]
Internuclear Distance R [\AA]
Exact UHF (Full-CI in basis)
PBE
r2SCAN
B3LYP
\textbf{(b)} Delocalization Catastrophe \& Post-Bifurcation Branch
Signed Functional Error $\Delta = E_{\mathrm{DFT}} - E_{\mathrm{UHF}}$ [mHa]
Internuclear Distance R [\AA]
PBE
r2SCAN
B3LYP
← Localized Branch ($\Delta \approx \mathrm{SIE}_{\mathrm{mol}}$)
← Delocalization Over-binding ($\Delta < 0$)
\textbf{(c)} Order Parameter for Spatial Symmetry Breaking
Proton Charge Asymmetry $|q(\mathrm{H}_a) - q(\mathrm{H}_b)|$
Internuclear Distance R [\AA]
PBE
r2SCAN
B3LYP
Symmetric (0.5 / 0.5)
Localized (1.0 / 0.0)